Analytics Questions (all 3 categories):

System Performance

1. How does syllabus length affect AI processing time?
2. How does AI processing latency change over time?
3. How long does the AI take on average to process a syllabus?

User Behavior

4. How frequently do students use the AI syllabus extractor?
5. Which input methods do students use most (PDF, DOCX, HTML, or pasted text)?
6. What percentage of extracted events do students actually save to their calendars?

Cost

7. How many tokens does Qwen generate per extraction request?
8. What input types result in the greatest token usage?
9. What is the estimated cost per request, and how does total cost change as usage increases?

Dashboard Planning:

Widget	Question it answers	Data required
Scatter plot: Syllabus Length vs. Processing Time	1	Syllabus length (characters), processing time (seconds)
Line chart: AI Latency Over Time	2	Request timestamp, processing time (seconds)
Summary metric: Average Processing Time	3	Processing time for each request
Line chart: AI Usage Over Time	4	Request timestamp, number of requests

Bar chart: Input Method Usage	5	Input method (PDF, DOCX, HTML, pasted text), request count
Pie chart: Calendar Save Rate	6	Number of events extracted, number of events saved
Summary metric: Average Generated Tokens	7	Generated tokens per request
Bar chart: Token Usage by Input Type	8	Input method, generated token count
Line chart: Estimated Cost Over Time + Summary Metric	9	Processing time, estimated computation cost per request, request timestamp

Data Science Thinking:

For our dashboard, we plan to use averages, trends over time, percentages, and comparisons across groups to analyze our AI system's performance, user behavior, and cost:

- We will use averages to calculate the mean processing time and the average number of tokens generated per request. This will help us understand how long the AI typically takes to process a syllabus and how many tokens it uses.
- We will use trends over time to track changes in processing latency, how frequently students use the extractor, and how estimated computational costs change as usage increases. This can help us identify patterns and determine whether the system becomes slower or more expensive over time.
- We will use percentages to measure the proportion of extracted events that students actually save to their calendars. This will help us understand how useful students find the AI-generated results.

- We will use comparisons across groups to see which input methods, such as PDF, DOCX, HTML, or pasted text, generate the most tokens. We will also use a scatter plot to examine the relationship between syllabus length and processing time. This will help us understand whether longer inputs require more processing time and resources.

Dashboard Screenshot:

****DONT FORGET in github submission comments: link to the dashboard!!**